

CHAITANYA S.

Applied AI Engineer | Generative AI | RAG | Agentic AI | AI Platform Engineering

414-396-8566 | chaitanya.aicareers@gmail.com

LinkedIn | GitHub | Hugging Face | Portfolio

Milwaukee, WI | Open to Remote / Relocation

PROFESSIONAL SUMMARY

Applied AI Engineer specializing in production grade LLM applications, Retrieval-Augmented Generation (RAG), GraphRAG, agentic AI systems, and document intelligence platforms built with Python, FastAPI, LangGraph, LangChain, LlamaIndex, vector databases, and LLM orchestration frameworks. Experienced designing grounded, citation backed AI workflows using hybrid retrieval, reranking, structured outputs, evaluation, guardrails, and human in the loop controls. Combines AI engineering with software engineering experience in FDA regulated pharmaceutical environments, including 21 CFR Part 11, GxP, and ALCOA+, to build secure, auditable, compliance aware AI systems.

CORE SKILLS

Generative AI & LLMs: RAG Architecture, LangChain, LlamaIndex, OpenAI API, GPT-4/4o, Hugging Face Transformers, Prompt Engineering, Structured Outputs, Hallucination Mitigation

Agentic AI & Automation: AI Agents, LangGraph, Multi Agent Systems, Model Context Protocol (MCP), Tool / Function Calling, Task Routing, Human in the Loop Approvals, Sandboxed Execution

Retrieval & NLP: Embeddings, Semantic Search, Semantic Chunking, Hybrid Search, BM25, GraphRAG, Dense Vector Retrieval, Metadata Filtering, Reranking, Citation Grounded Generation

AI Evaluation & Safety: Phoenix, LangSmith, Groundedness Evaluation, Retrieval Precision, NeMo Guardrails, Llama Guard, Prompt Injection Defense, Audit Logging

Backend & Infrastructure: Python, SQL, FastAPI, REST APIs, Async Architecture, Pydantic, Streamlit, Pandas, NumPy, PostgreSQL, ChromaDB, Pinecone, Docker, AWS EC2/S3, CI/CD, Git, GitHub

Security & Compliance: RBAC, FDA 21 CFR Part 11, GxP, ALCOA+, Regulatory Document Workflows, SOP, CAPA, Deviation Workflows, Validation Oriented Systems

PROFESSIONAL EXPERIENCE

Founder & Lead AI Engineer / Architect

June 2026 - Present

SPACTR AI Labs LLC - Milwaukee, WI (Remote)

Tech Stack: Python, FastAPI, LangGraph, MCP, LangChain, LlamaIndex, ChromaDB, Pinecone, NeMo Guardrails, Llama Guard, Streamlit, Docker, AWS EC2/S3, OpenAI API, Hugging Face

Enterprise AI Platforms Developed

PharmaAI Platform - Regulatory, Summarization & Compliance Intelligence System

- Architected an end to end RAG pipeline using LlamaIndex and custom ingestion workflows covering semantic chunking, vector indexing, reranking, and citation generation for FDA guidance documents and regulatory PDFs.
- Extended retrieval precision using hybrid search, dense vectors, BM25, and GraphRAG for relationship aware query resolution across regulatory, SOP, CAPA, and deviation workflows, evaluating groundedness and latency through Phoenix, LangSmith, and custom evaluation suites.
- Engineered a compliance focused QA assistant integrating NeMo Guardrails, Llama Guard, prompt injection defense, structured outputs, mandatory inline citations, and evidence backed answer generation.
- Consolidated retrieval, summarization, compliance QA, and document intelligence workflows behind shared asynchronous FastAPI services with RBAC, audit logging, and validation review controls.

Agentic AI Orchestration Platform - Local-First & Private-Cloud Multi Agent System

- Built a local first, privacy focused multi-agent orchestration platform on LangGraph using deterministic state machine control, modular agents, dynamic task routing, and execution planning.
- Implemented tool/function calling with human in the loop approval gates and sandboxed execution runtimes so operators explicitly authorize sensitive AI actions before execution.
- Integrated external tools through Model Context Protocol (MCP) servers, establishing a standardized interface for agents to query local files, PostgreSQL databases, REST APIs, and external systems.
- Developed a database backed wait-state pattern that serializes agent state and memory to PostgreSQL during approval pauses and resumes execution without context loss.

AI/ML Engineer

May 2024 - May 2026

SolutionsMax Technology Services Inc - Sacramento, CA (Remote)

Tech Stack: Python, FastAPI, OpenAI API, LangChain, Pinecone, PostgreSQL, Docker, AWS EC2/S3, REST APIs, CI/CD

- Shipped production LLM-powered summarization, entity extraction, and knowledge retrieval features for enterprise regulatory documentation, processing 50,000+ pages of compliance text.
- Built retrieval based AI capabilities into enterprise backend services through RESTful FastAPI microservices, maintaining sub 300 ms API response latency under high-volume query loads.
- Designed and optimized vector search indexing pipelines using Pinecone and PostgreSQL, including custom chunking, embedding workflows, and metadata filtering to improve retrieval accuracy.
- Implemented prompt engineering patterns, Pydantic based structured JSON outputs, and validation guardrails to reduce hallucination risk across automated compliance workflows.
- Built preprocessing and validation workflows that converted unstructured regulatory content into retrieval ready chunks and Pydantic validated structured outputs for downstream AI services.
- Containerized AI microservices using Docker and deployed scalable API endpoints on AWS EC2 and S3 with CI/CD practices for repeatable releases.
- Partnered cross functionally with product and engineering teams to design automated document intelligence workflows, reducing manual regulatory review cycles by 35%.

Software Engineer - Regulated Data Systems

June 2020 - July 2023

SolutionsMax Technology Services Pvt Ltd - Visakhapatnam, India

Tech Stack: Python, SQL, PostgreSQL, ETL Pipelines, Data Modeling, QA Automation

- Engineered Python and SQL data and reporting pipelines for FDA regulated systems aligned with 21 CFR Part 11 and ALCOA+ data integrity standards.
- Designed relational data models and automated ETL workflows that improved data traceability and auditability across pharmaceutical product lifecycle systems.
- Built validation, transformation, and reporting workflows for structured enterprise datasets supporting regulatory, quality, and compliance processes.
- Automated validation and reporting processes, reducing manual QA testing overhead by 40% and strengthening audit readiness across regulated datasets.
- Authored functional specifications and technical design documents mapping regulatory mandates and business requirements into implementation ready engineering deliverables.